

Project

1. A SWITCH OFF=0 ON=1

GATE Z

B TIMER OFF=0 ON=1

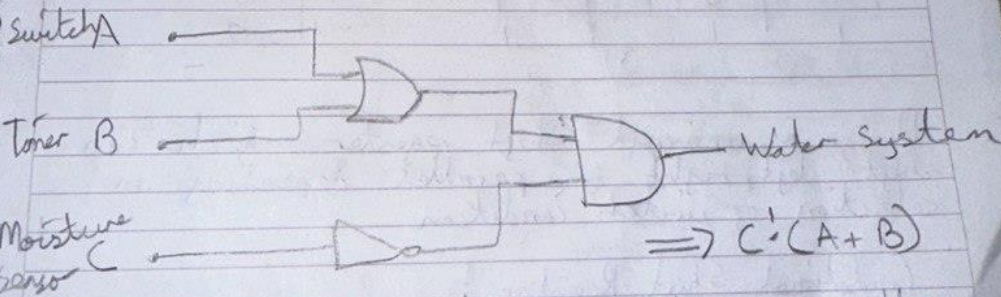
GATE Y

C MOISTURE SENSOR DRY=0 WET=1

GATE X

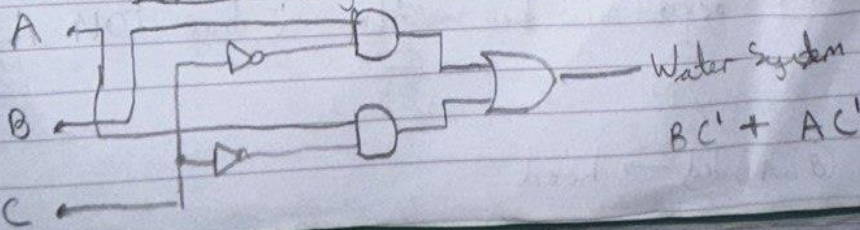
- a. Gate X = AND gate
Y = NOT gate
Z = OR gate

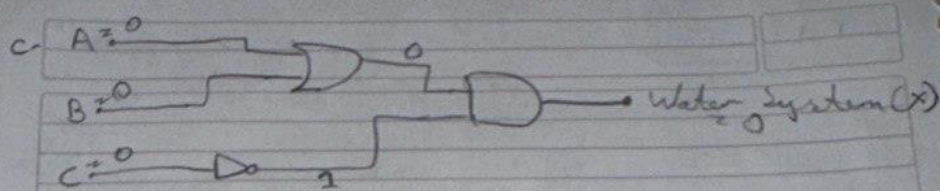
- b. If soil is wet, the output of moisture sensor is 1 but system MUST only activate when the soil is dry



Switch is the manual form of turning on the system and the timer is automatic. We use an OR gate because both of them don't have to be turned "ON" (or 1) for the system to work.

Alternatively:





If it is a dry day, $C = 0$ (motion sensor)
 If A and B are 0

The water system is 0

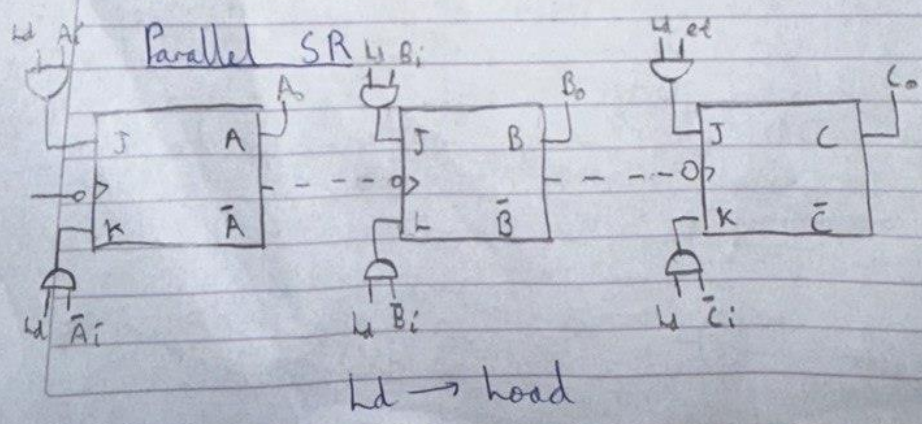
No, it won't be activated

d.

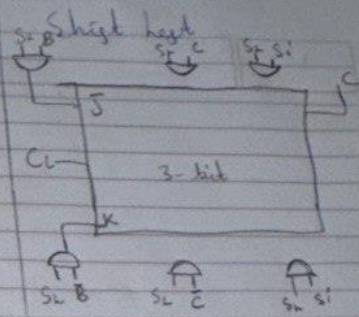
Switch (A)	Timer (B)	Motion Sensor (C)	Water System (x)
0	0	0	0
0	1	0	1
0	0	1	0
0	1	1	0
1	0	0	1
1	1	0	1
1	0	1	0
1	1	1	0

2. Design a universal shift register which can shift left, right or parallel depending on selection of switch condition

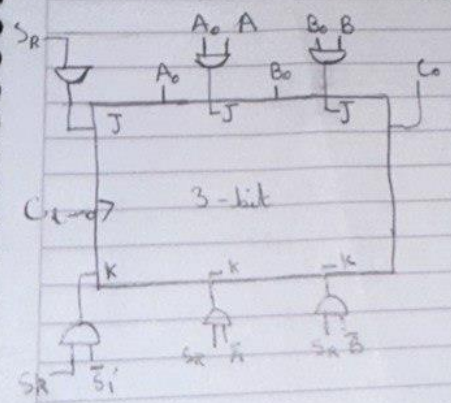
Conditional Shift Register:



Clock	Load	Ai	Bi	Bi	Ao	Bo	Co
R	X	X	X	X	0	1	0
F	2	1	0	0	1	0	1
F	2	0	0	0	1	0	1
R	2	1	1	1	1	0	1
R	2	1	1	1	1	0	1
F	2	0	1	1	1	0	1
F	2	1	1	1	1	1	1

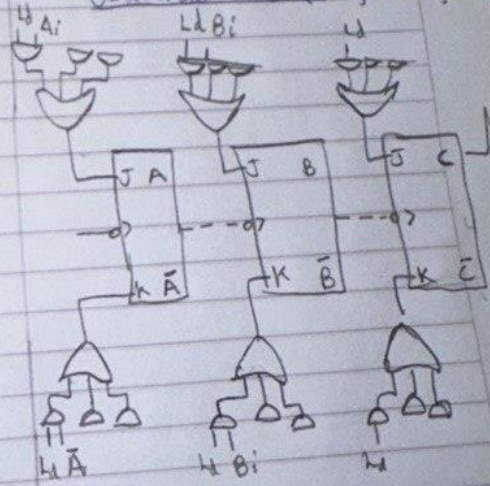


Shift Right Register



Cl	Si	Si	A	B	C
X	X	X	0	1	0
2	1	0	0	1	0
2	0	0	0	1	0
2	1	0	1	0	0
2	1	1	1	0	0
2	1	1	0	0	1
2	0	1	0	0	1
2	1	1	0	1	1

Universal SR Shift Register



Clock	Se	Se	Ao	Bo	Co
X	X	X	0	1	0
R	2	1	0	1	0
F	2	0	0	1	0
F	2	1	0	0	1
R	2	1	1	0	1
R	2	1	1	0	0
F	2	0	1	0	0
F	2	1	1	1	0